MILITARY SPECIFICATION

BRUSHES, ELECTRICAL CONTACT, RECTANGULAR, STANDARD DESIGNS AND DIMENSIONS FOR

To all activities:

1. The following page of this standard has been revised and supersedes the page listed:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
9	Reprinted without change		
10	26 July 1961	10	8 December 1960

2. Retain this cover page and insert before the table of contents of this standard.

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Round type, stamped.

Round type, machined.

Eared type.

4.2.4.2 *Dimensions.*

4.2.4.2.1 *Round type, stamped.* The dimensions shall conform to table XI. The terminal shall conform to figure 16.

4.2.4.2.2 *Round type, machined.* The dimensions shall conform to table XII. The terminal shall conform to figure 17.

TABLE XI. *Stamped, round type terimnals.*

Designation	D1	D2	T1
	Inch	*Inch*	*Inch*
0120903	3/16	9/64	3/64
0161105	1/4	11/16	5/16
0201504	5/16	15/64	1/16
0241604	3/8	1/4	1/16
0282207	7/16	11/32	7/64
0262608	9/16	7/16	1/8
0483208	3/4	1/2	1/8

NOTE.—The plan for setting up the designations under table XI is as follows: At the left are the type letters. Each pair of numbers designates the number of sixty-fourths of an inch in the appropriate dimension. The pairs are lined up in the same order as the columns are in the table. These designations correspond with the NEMA full designations in ASA Publication No. C64. 1-1956, NEMA Publication No. CB 1-1956.

TABLE XII. *Machined, round type terminals.*

Designation	D1	D2	T1	T2
	Inch	*Inch*	*Inch*	*Inch*
OM 1411-0603	7/32	11/64	3/32	3/64
OM 1613-0604	1/4	13/64	3/32	1/16
OM 1816-0806	9/32	1/4	1/8	1/16
OM 2018-0804	5/16	9/32	1/8	1/16
OM 2218-0804	11/32	9/32	1/8	1/16
OM 3222-0804	1/2	11/32	1/8	1/16
OM 3626-0804	9/16	13/32	1/8	1/16

4.2.4.2.3 *Eared type.* The dimensions shall conform to table XIII. In no case shall the length of the terminal lug ears exceed the solid compressed length of the spring (see 4.2.5). The terminal shall conform to figure 18.

TABLE XIII. *Eared type terminals.*

Designation	D1	W1	W2	We	Le
	Inch	*Inch*	*Inch*	*Inch*	*Inch*
U161412-0512	1/4	7/32	3/16	5/64	3/16
U181612-0812	9/32	1/4	3/16	1/8	3/16
U221814-1414	11/16	9/32	7/32	7/32	7/32
U302420-1616	15/32	3/8	5/16	1/4	1/4
U383228-1624	19/32	1/2	7/16	1/4	3/8

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TABLE XIV. *Standard sizes of brushes having both shunts and springs.*

Designation	Dimensions of brush material (in 1/16 inch)			Minimum permissible pressure (ounces)		Maximum permissible pressure (ounces)	Compressed length of spring (in 1/16 inch)	
	Thickness	Width	Length	Condition 1	Condition 2	Condition 1	Condition 1	Condition 2
030412	3	4	12	4.0	3.4	4.5	12	18
040512	4	5	12	6.7	5.7	7.5	12	18
040616	4	6	16	8.1	6.9	9.0	12	18
040716	4	7	16	9.4	8.0	10.5	12	18
040816	4	8	16	10.8	9.2	12.0	16	24
041020	4	10	20	13.5	11.5	15.0	20	30

NOTES:

1. The above designation is based on expressing in one-sixteenths of an inch, the thickness, width and length in that order.
2. Condition 1 represents the condition when the brush is new. Condition 2 represents the condition when the brush has undergone 100 percent wear. (For the purposes of this table, the following assumptions were made in deriving the value for condition 2: A length of head equal to 11 percent of the total length of the brush. The point of 100 percent wear being at a distance from the contact surface equal to 0.65 multiplied by the remainder obtained by subtracting the length of the head from the total length of the brush.)
3. The distance between the commutator or collector ring and the bottom of the brush holder cap or bottom of the recess in the cap equal to twice the length of the brush when new.
4. Spring pressure is specified on the following basis:
   a. A maximum pressure of 6.0 pounds per square inch when the brush is new.
   b. A minimum pressure of 5.4 pounds per square inch when the brush is new.
   c. A minimum pressure of 4.8 pounds per square inch when the brush has undergone 100 percent wear.

4.2.4.2.4 *Tolerances.* The tolerances of the dimensions appearing in tables XI, XII and XIII shall be as follows:

Le—plus or minus 1/64 inch.

All others—plus 0.000 inch, minus 0.008 inch.

4.2.5 *Springs.* Springs shall be of the positive gradient, compression, helical wire type. There shall be two closed turns at each end of the spring.

4.2.6 *Marking.* Each brush shall be marked with the following information:

(a) Manufacturers' grade designation. If possible, this marking shall be placed above the point of 100 percent wear so that the information is still available when the brush has worn to that point.

(b) Indication of points of 75 and 100 percent wear. The wear points shall be so marked that there is a continuous marking between the two points of wear along the length of the brush.

4.2.7 *Standard sizes.* Standard sizes for rectangular brushes with shunts and springs shall conform to table XIV. The shunts for these brushes shall conform to table XV.

TABLE XV.

Shunts for standard rectangular brushes with springs.

Brush designation	Minimum permissible shunt size where brush current density is 60 amp./sq. in. or less (see note)	Minimum permissible shunt size where brush current density is 100 amp./sq. in. or less but greater than 60 amp./sq. in. (see note)
030412	7-15-002	3-50-002
040512	7-15-002	7-11-003
040616	3-50-002	7-40-002
040716	7-11-003	7-27-003
040816	7-13-003	7-27-003
041020	7-15-003	7-27-003

NOTE.—In these shunt designations, the first figure represents the number of members, the second figure represents the number of strands in each member, and the third figure represents the nominal diameter of strands in inches.

4.3 Type III, rectangular, with springs but without shunts.

4.3.1 *Dimensions and tolerances.* The requirements of 4.2.1, 4.2.2, and 4.2.5 shall apply.

4.3.2 *Marking.* Each brush shall be marked with the manufacturer's grade designation.

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